Seasonality

References:

Tsay (2010), Sections 2.8; Enders (2014), Section 2.11

1 Introduction

- Many economic and financial time series exhibit some form of *seasonality*, which is a regular pattern of changes that repeats over certain time periods.
- Seasonal variation could arise from various factors, including weather (e.g., agriculture, construction, tourism), holidays (e.g., retail sales), payment and financial reporting schedules, etc.
- Why do we care about seasonal variation
  - Forecasts that ignore strong seasonal patterns will have large variance.
  - Regular seasonal movement may overshadow the true underlying economic mechanisms and relationships.
- The method of removing the seasonal component of a time series is referred to as *seasonal adjustment* or *deseasonalization*. A series that has its seasonal component removed is called *seasonally adjusted* or *deseasonalized* data.
- Caution is needed even in the use of deseasonalized data published by a government agency. A standardized *deseasonalization* procedure may not work the best for a specific series.
- Two approaches to handle seasonality
  - Remove seasonality first and then apply $ARMA$ models using the Box-Jenkins techniques
  - Jointly identify and estimate seasonal models

2 Seasonal Differencing

- Recall that $\Delta y_t = y_t - y_{t-1} = (1-L)\, y_t$, which is referred to as the *regular (or nonseasonal) differencing*. Also, $\Delta^2 y_t = (1-L)^2\, y_t = y_t - 2y_{t-1} + y_{t-2}$.

- *Seasonal differencing* refers to the procedure of taking differences between observations in the same season of consecutive years, instead of consecutive periods. Let S be the number of time periods until the seasonal pattern repeats. Then a seasonal difference is defined by $\Delta_S y_t = y_t - y_{t-S} = \left(1 - L^S\right) y_t$.

  - For monthly data, $S = 12$ (months per year). A seasonal difference is

$$\Delta_{12} y_t = \left(1 - L^{12}\right) y_t = y_t - y_{t-12}.$$

  - For quarterly data, $S = 4$ (quarters per year). A seasonal difference is

$$\Delta_4 y_t = \left(1 - L^4\right) y_t = y_t - y_{t-4}.$$

- The second-order seasonal difference is

$$\begin{aligned} \Delta_S^2 y_t &= \Delta_S \left(y_t - y_{t-S}\right) = \Delta_S y_t - \Delta_S y_{t-S} \\ &= \left(y_t - y_{t-S}\right) - \left(y_{t-S} - y_{t-2S}\right) \\ &= y_t - 2y_{t-S} + y_{t-2S}. \end{aligned}$$

- Combining seasonal difference with first difference:

$$(1-L)\left(1 - L^S\right) y_t = \left(1 - L - L^S + L^{S+1}\right) y_t = y_t - y_{t-1} - y_{t-S} + y_{t-S+1}.$$

- Does the order of differencing matter?

- Combining the two types of differencing in general yields $\Delta^d \Delta_S^D$, where d is the number of nonseasonal differences, and D is the number of seasonal differences.

3 Seasonal ARIMA Models

- Seasonal effects can be modeled stochastically by the seasonal $ARMA$ models. Pure seasonal $ARMA\left(P,Q\right)_S$ models with period S has the form

$$\Phi\left(L^S\right)\left(y_t-\mu\right)=\Theta\left(L^S\right)\varepsilon_t, \tag{1}$$

  where $\varepsilon_t\sim WN\left(0,\sigma^2\right)$,

$$\Phi\left(L^S\right)=1-\Phi_1L^S-\Phi_2L^{2S}-\ldots-\Phi_PL^{PS}, \tag{2}$$
$$\Theta\left(L^S\right)=1+\Theta_1L^S+\Theta_2L^{2S}+\ldots+\Theta_QL^{QS}. \tag{3}$$

- Similar to pure $ARMA$ process, the pure seasonal $ARMA$ process $\{y_t\}$ given by (1) is stationary and invertible if all the roots of $\Phi\left(z\right)=0$ and $\Theta\left(z\right)=0$ lie outside of the unit circle, respectively.

Example 3.1. Seasonal $ARMA\left(1,0\right)_4$ has the form

$$\left(1-\Phi_1L^4\right)y_t=\varepsilon_t,\quad \text{i.e.,}\quad y_t=\Phi_1y_{t-4}+\varepsilon_t.$$

Example 3.2. Seasonal $ARMA\left(1,1\right)_{12}$ has the form

$$\left(1-\Phi_1L^{12}\right)y_t=\left(1+\Theta_1L^{12}\right)\varepsilon_t,$$

namely,

$$y_t=\Phi_1y_{t-12}+\varepsilon_t+\Theta_1\varepsilon_{t-12}.$$

- $\{y_t\}$ is a pure seasonal $ARIMA\left(P,D,Q\right)_S$ process if $x_t=\left(1-L^S\right)^Dy_t$ is a pure $ARMA\left(P,Q\right)_S$ process: $\Phi\left(L^S\right)x_t=\Theta\left(L^S\right)\varepsilon_t$.

- In most applications, D is not greater than one, and P and Q are typically less than three.

Exercise 3.1. Find the ACF of the process $y_t=\varepsilon_t-0.3\varepsilon_{t-4}$, where $\varepsilon_t\sim WN\left(0,\sigma^2\right)$.

Exercise 3.2. Find the ACF of the process $y_t = 0.7y_{t-4} + \varepsilon_t$, where $\varepsilon_t \sim WN\left(0, \sigma^2\right)$.

- More generally, seasonal $ARIMA$ (or $SARIMA$) models combine both non-seasonal and seasonal factors in a multiplicative manner. In specific, $ARIMA\left(p,d,q\right)\left(P,D,Q\right)_S$ models have the form

$$\phi\left(L\right)\Phi\left(L^S\right)\left(1-L\right)^d\left(1-L^S\right)^D\left(y_t-\mu\right)=\theta\left(L\right)\Theta\left(L^S\right)\varepsilon_t,$$

or equivalently,

$$\phi\left(L\right)\Phi\left(L^S\right)\Delta^d\Delta_S^D\left(y_t-\mu\right)=\theta\left(L\right)\Theta\left(L^S\right)\varepsilon_t,$$

where the non-seasonal AR and MA components are

$$\phi\left(L\right)=1-\phi_1L-\ldots-\phi_pL^p,\quad \theta\left(L\right)=1+\theta_1L+\ldots+\theta_qL^q,$$

the seasonal AR and MA components are given by (2) and (3), respectively,

  - $p=$ order of non-seasonal AR terms
  - $d=$ order of non-seasonal differencing
  - $q=$ order of non-seasonal MA terms
  - $P=$ order of seasonal AR terms
  - $D=$ order of seasonal differencing
  - $Q=$ order of seasonal MA terms
  - $S=$ seasonal period (e.g., number of seasons in a year)

Example 3.3. $ARIMA\left(0,0,1\right)\times\left(0,0,1\right)_{12}$:

$$\begin{aligned}y_t-\mu&=\left(1+\theta_1L\right)\left(1+\Theta L^{12}\right)\varepsilon_t\\&=1+\theta_1L+\Theta_1L^{12}+\theta_1\Theta_1L^{13}\\&=\varepsilon_t+\theta_1\varepsilon_{t-1}+\Theta_1\varepsilon_{t-12}+\theta_1\Theta_1\varepsilon_{t-13}.\end{aligned}$$

Verify that this process will only have non-zero autocorrelations at lags 1, 11, 12, and 13. In particular, notice that the covariance between y_t and y_{t-11} is given by

$$E\left(\varepsilon_t + \theta_1\varepsilon_{t-1} + \Theta_1\varepsilon_{t-12} + \theta_1\Theta_1\varepsilon_{t-13}\right)\left(\varepsilon_{t-11} + \theta_1\varepsilon_{t-12} + \Theta_1\varepsilon_{t-23} + \theta_1\Theta_1\varepsilon_{t-24}\right),$$

and hence the lag-11 autocorrelation is non-zero.

Example 3.4. $SARIMA\left(1,0,0\right)\times\left(1,0,0\right)_{12}$:

$$\left(1-\phi_1 L\right)\left(1-\Phi_1 L^{12}\right)\left(y_t-\mu\right)=\varepsilon_t.$$

Let $x_t = y_t - \mu$. Then

$$\left(1-\phi_1 L-\Phi_1 L^{12}+\phi_1\Phi_1 L^{13}\right)x_t=\varepsilon_t$$

$$x_t=\phi_1 x_{t-1}+\Phi_1 x_{t-12}-\phi_1\Phi_1 x_{t-13}+\varepsilon_t.$$

What will the ACF and PACF of y_t look like?

- The $SARMA$ model

$$\phi\left(L\right)\Phi\left(L^S\right)\left(y_t-\mu\right)=\theta\left(L\right)\Theta\left(L^S\right)\varepsilon_t,$$

  is stationary if all the roots of $\phi\left(z\right)=0$ and $\Phi\left(z\right)=0$ lie outside of the unit circle. It will be invertible if all the roots of $\theta\left(z\right)=0$ and $\Theta\left(z\right)=0$ lie outside of the unit circle.

- The multiplicative seasonal models are more parsimonious than non-multiplicative (or additive) models. To see this, compare

$$y_t=\left(1+\theta L\right)\left(1+\Theta L^S\right)\varepsilon_t=\varepsilon_t+\theta\varepsilon_{t-1}+\Theta\varepsilon_{t-S}+\theta\Theta\varepsilon_{t-S-1},$$

$$y_t=\left(1+\theta L+\Theta L^S+\gamma L^{S+1}\right)\varepsilon_t=\varepsilon_t+\theta\varepsilon_{t-1}+\Theta\varepsilon_{t-S}+\gamma\varepsilon_{t-S-1}.$$

  The first equation requires estimating two parameters, whereas the second equation requires estimating three parameters.

4 Identifying Seasonal ARIMA Models

- Basic steps to model seasonal data

  1. Draw a time series plot of the data.
  2. Do differencing if necessary.
     - If there is a linear trend but no clear seasonality: take a first difference? (stochastic trend vs. deterministic trend?) If there is a non-linear trend, consider a transformation (such as taking logs) of the data before differencing.
     - If there is seasonality and but no trend: take a seasonal difference (i.e., lag-S difference)? (stochastic vs. deterministic seasonality?)
     - If there are both trend and seasonality: take a seasonal difference first and re-examine the trend; if a trend remains, then also consider taking a first difference.
  3. Examine the ACF and PACF of the (differenced) data.
     - Examine low lags to judge non-seasonal terms. Spikes in the ACF (PACF) at low lags suggest non-seasonal MA (AR) terms.
     - Examine the patterns across lags that are multiples of S to judge seasonal terms. Interpret the implications of ACF/PACF in the same way as you do for the lower lags.
  4. Estimate reasonable candidate models on the basis of Step 3.
  5. Compare AIC/BIC to select the most appropriate model and perform diagnostic checking (time series plot, ACF, Box-Pierce test, etc.).

- Deterministic vs. stochastic seasonality

  - If a seasonal pattern is stable over time (close to a deterministic function), one may consider regression on seasonal dummy variables to model seasonality. For example, when $S = 4$,

$$y_t = \alpha_0 + \alpha_1 D_{1t} + \alpha_2 D_{2t} + \alpha_3 D_{3t} + \varepsilon_t,$$

or

$$y_t = \alpha_1 D_{1t} + \alpha_2 D_{2t} + \alpha_3 D_{3t} + \alpha_4 D_{4t} + \varepsilon_t.$$

In general,

$$y_t = \alpha_0 + \sum_{s=1}^{S-1} \alpha_s D_{st} + \varepsilon_t. \tag{4}$$

– Note that (4) implies that y_t has a seasonally shifting mean, and hence y_t is nonstationary.

– Given finite samples observed in practice, the deterministic seasonal and seasonal unit root processes are empirically similar. For example, consider the following two competing specifications:

$$y_{s\tau} = \gamma_s + \varepsilon_{s\tau}, \tag{5}$$

$$\Delta_S y_{s\tau} = \left(1 - \Theta L^S\right) \varepsilon_{s\tau}, \tag{6}$$

for $s = 1, 2, \ldots, S$ and $\tau = 1, 2, \ldots, T_\tau$, where $y_{s\tau}$ is the τth observation in season s, and for notational simplicity we assume $T = ST_\tau$. Note that the series

$$y_1, y_2, \ldots, y_S, y_{S+1}, y_{S+2}, \ldots, y_{2S}, y_{2S+1}, \ldots$$

is now denoted using the double subscripts

$$y_{11}, y_{21}, \ldots, y_{S1}, y_{12}, y_{22}, \ldots, y_{S2}, y_{13}, \ldots$$

It is not difficult to verify that (6) implies

$$y_{s\tau} = y_{s,\tau-1} + \left(1 - \Theta L^S\right) \varepsilon_{s\tau} = y_{s0} + \sum_{i=1}^{\tau} \left(1 - \Theta L^S\right) \varepsilon_{si}.$$

If $\Theta = 1$, the above equation becomes

$$y_{s\tau} = y_{s0} + \sum_{i=1}^{\tau} (\varepsilon_{si} - \varepsilon_{s,i-1}) = y_{s0} - \varepsilon_{s0} + \varepsilon_{s\tau},$$

which is identical to the deterministic seasonality equation, (5), if $y_{s0} - \varepsilon_{s0} = \gamma_s$.

– The above result also holds when $\varepsilon_{s\tau}$ is replaced with more general stationary $ARMA$ processes.

– Note that Δ_S and the MA operator $\left(1 - \Theta L^S\right)$ in (6) cancel out when $\Theta = 1$. The logic can be extended to the "near cancellation" case in which Θ is close to unity and it is difficult to discriminate between seasonal unit root process and deterministic seasonal process especially for small sample size. (See, Tsay (2010), Example 2.4).